

Website vs Excel Calculation Comparison

South/South-West hotel cases: 50 to 250 rooms. Excel side follows formulas/constants visible in the uploaded workbook; website side uses the current backend calculation engine with the same main room, tariff, fuel and profile assumptions. Sustainable tourism premium scenario is set to No to match the uploaded workbook.

Technical and CAPEX Outputs

Rooms	Web RT	Excel RT	Web kW	Excel kW	Web CAPEX	Excel CAPEX	Web t/mo	Excel t/mo
50	250	250	300	300	85.22M	146.55M	313.1	313.1
100	350	350	400	400	116.76M	200.64M	417.5	417.5
150	450	450	500	500	148.29M	254.72M	521.9	521.9
200	650	650	800	800	224.14M	385.57M	835	835
250	750	750	1000	1000	268.45M	462.34M	1,043.7	1,043.7

Financial Outputs

Rooms	Web NPV	Excel NPV	Web IRR	Excel IRR	Web PB	Excel PB	Web DPB	Excel DPB
50	490.24M	337.08M	60.56%	45.84%	1.85	2.13	2.17	2.65
100	432.96M	445.91M	44.12%	44.26%	2.68	2.22	3.19	2.77
150	375.69M	554.73M	34.61%	43.35%	3.6	2.27	4.36	2.84
200	878.38M	912.66M	45.81%	46.25%	2.56	2.12	3.04	2.64
250	1.11B	1.17B	47.65%	48.42%	2.44	2.03	2.89	2.5

Decision and NPV Difference

Rooms	Website	Excel	NPV Diff Web-Excel	NPV Diff %
50	FEASIBLE	FEASIBLE	153.16M	45.4%
100	FEASIBLE	FEASIBLE	-12.94M	-2.9%
150	FEASIBLE	FEASIBLE	-179.04M	-32.3%
200	FEASIBLE	FEASIBLE	-34.28M	-3.8%
250	FEASIBLE	FEASIBLE	-53.34M	-4.6%

Key observation

The two outputs differ mainly because the current website engine and the uploaded Excel workbook treat some downstream cash-flow and capital-cost details differently. The uploaded workbook includes the expanded CAPEX build-up with tax uplift and uses the workbook cash-flow rows for export and feasibility indicators. This comparison is generated as a separate file and does not modify the application code.